

# 1. HIGH-LEVEL DESIGN (HLD)

## 1. System Overview

This system is designed to simulate an intelligent customer support assistant capable of answering user queries based on a predefined knowledge base.

The system uses a **Retrieval-Augmented Generation (RAG)** approach to ensure that responses are grounded in factual data rather than generated purely from the language model.

### Problem Statement

Traditional chatbots rely solely on LLMs, which may generate incorrect or hallucinated responses.

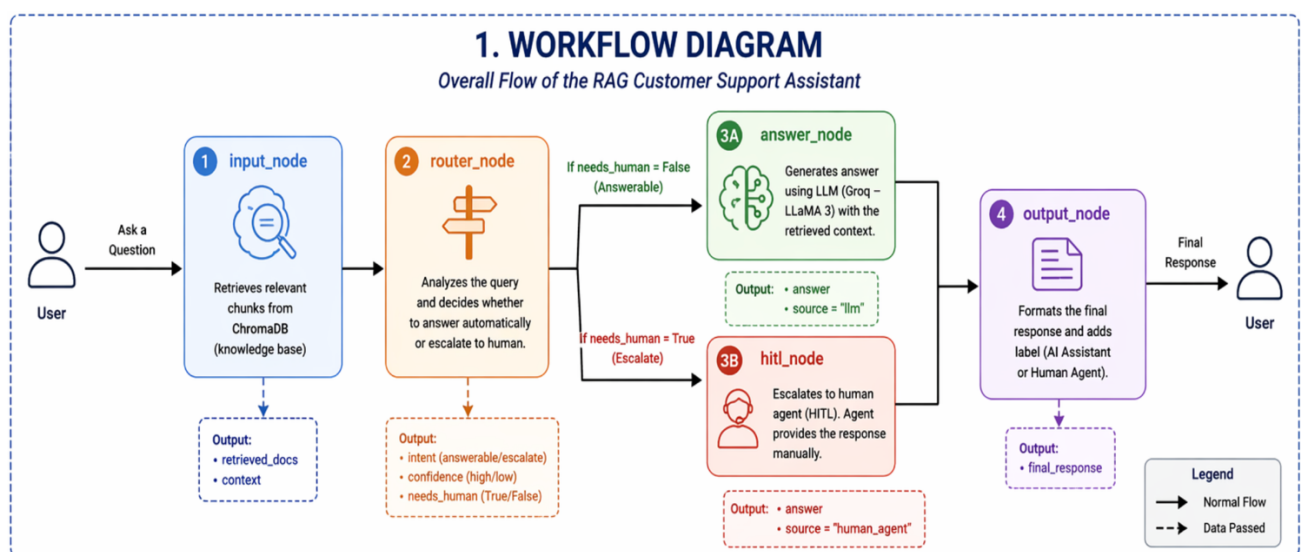
This project addresses this issue by:

- Retrieving relevant information from a trusted source (PDF)- in this project, I am using “TechEase Customer Support” knowledge base
- Generating responses only using that information

### Scope

- Answer customer queries from PDF knowledge base
- Perform intelligent routing
- Escalate complex queries to human agents

## 2. Architecture/Work Flow Diagram



The system consists of the following layers:

## **1. User Interface**

- CLI-based interaction (main.py)
- Accepts user queries

## **2. Document Processing Layer**

- Loads PDF
- Splits into chunks
- Converts into embeddings

## **3. Vector Storage Layer**

- Stores embeddings in ChromaDB
- Enables semantic search

## **4. Retrieval Layer**

- Finds top relevant chunks using similarity search

## **5. LLM Processing Layer**

- Uses Groq LLaMA model
- Generates final answer using retrieved context

## **6. Workflow Orchestration (LangGraph)**

- Controls flow between nodes
- Implements decision-making logic

## **7. Routing Layer**

- Decides:
  - Answer with AI
  - Escalate to human

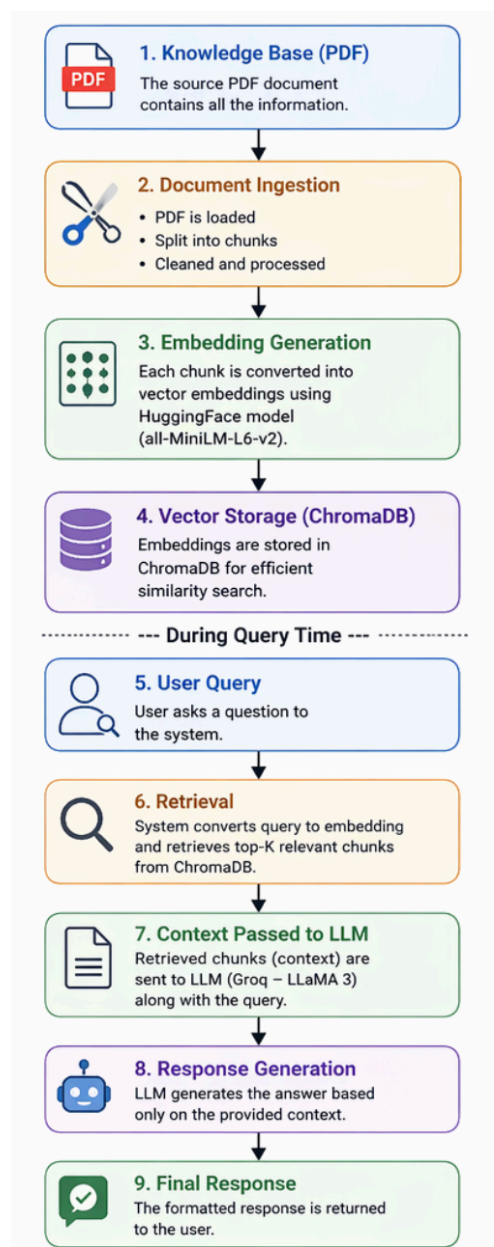
## **8. HITL System**

- Handles complex or sensitive queries
- Human agent provides final response

### 3. Components

- Document Loader: Loads PDF
- Chunking Module: Splits text into chunks
- Embedding Model: Converts text into vectors
- Vector Database (ChromaDB): Stores embeddings
- Retriever: Fetches relevant chunks
- LLM (Groq LLaMA 3): Generates answers
- LangGraph: Controls workflow
- HITL Module: Handles escalation

### 4. Data Flow



## **5. Technology Choices**

- ChromaDB: Lightweight vector DB
- HuggingFace Embeddings: Free and local
- Groq LLM: Fast inference
- LangGraph: Workflow control

## **6. Scalability**

- Supports large PDFs via chunking
- Vector DB enables fast search
- Can be extended to multi-user system